1. Ingestion Setup

Because raw files were stored in GitHub via Git LFS, they were pulled directly into memory using requests and io.BytesIO:

- Wages (2025 OEWS): Read the "All Areas" tab (statewide + 13 LWDAs).
- Demographics (Maryland Planning / Census): Read the "Regional_Share" tab, skipping two header banner rows (header=2).

2. Wage Data Cleaning

- Removed hourly metrics: Dropped all columns starting with "Hourly" to focus only on annual pay.
- Standardized column headers: Trimmed whitespace, converted to lowercase, and replaced consecutive spaces with single underscores.
- Handled missing data: Used a targeted drop only if both mean and median annual wages were missing. (The 96 missing employment counts and 30 missing experience values reflect standard state data suppression for tiny sample sizes to protect privacy. Core median wages have zero missing values).
- Occupation hierarchy flags: Added boolean flags (is_total, is_major_group, is_detailed) to distinguish overall totals (00-0000) and 2-digit industry groups (0000) from 6-digit jobs to prevent double-counting in regressions and t-tests.
- Type casting: Converted employment counts and annual wage metrics to numeric (float/int).

3. Demographics Data Cleaning and Feature Engineering

- Filtered population rows: Retained only "Jurisdiction," "Sex," "Race," and "2025" counts, keeping rows where Sex == 'Total' to exclude gender-split subsets.
- LWDA geographic mapping: Mapped Maryland's 24 jurisdictions into the 13 official LWDAs (e.g., grouping Allegany, Garrett, and Washington into Western Maryland).
- Pivoted race categories: Filtered for "Total Population" and "Total Racial and Ethnic Minority Population **" and pivoted them into separate columns summed by LWDA.
- Engineered demographics metric:

$$minority_pct = \left(\frac{minority_population}{total_population}\right) \times 100$$

4. Merging for Modeling

- Joined df_clean and df_lwda with area_name.
- Confirmed all 14 geographic areas (13 LWDAs + Statewide) merged cleanly with 0 missing demographic values.

Reproducibility Steps:

1. Run cells sequentially in 01_Ingestion.ipynb in Colab or Jupyter.
2. Ensure libraries are imported and installed.
3. Execute Part 1a to produce MD_OEWS_Cleaned.csv.
4. Execute Part 1b to produce MD_LWDA_Demographics_Summary.csv.
5. Execute Part 1c to produce MD_Wage_Demographics_Master.csv for modeling.